Learning Video-Audio Alignment Through Multimodality Representation

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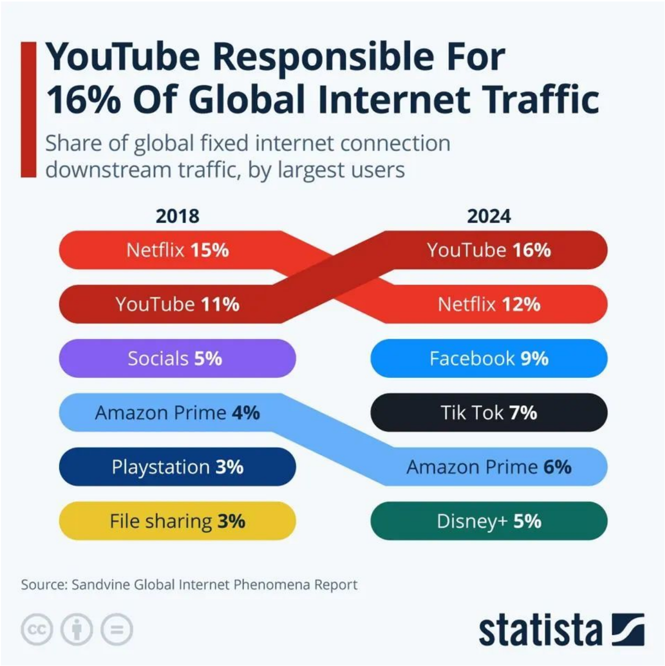
Introduction

Data is being sent shared, distributed, and sent through the internet at inconceivable rates

- Audio and video is extremely popular throughout the web!

Video is, more or less, a majority of internet traffic

- In 2024, Forbes reported that just YouTube made up 16% of all internet traffic[1]!



[2]

[1][2] Katharina Buchholz "YouTube Responsible For 16% Of Global Internet Traffic". https://www.forbes.com/sites/katharinabuchholz/2025/04/25/youtube-responsible-for-16-of-global-internet-traffic/

Motivation

Data can be corrupted and misrepresented when shared through the internet

- For audio/video, this can be noise in the respective channels or **misalignment**
- Non-intentionally or intentionally

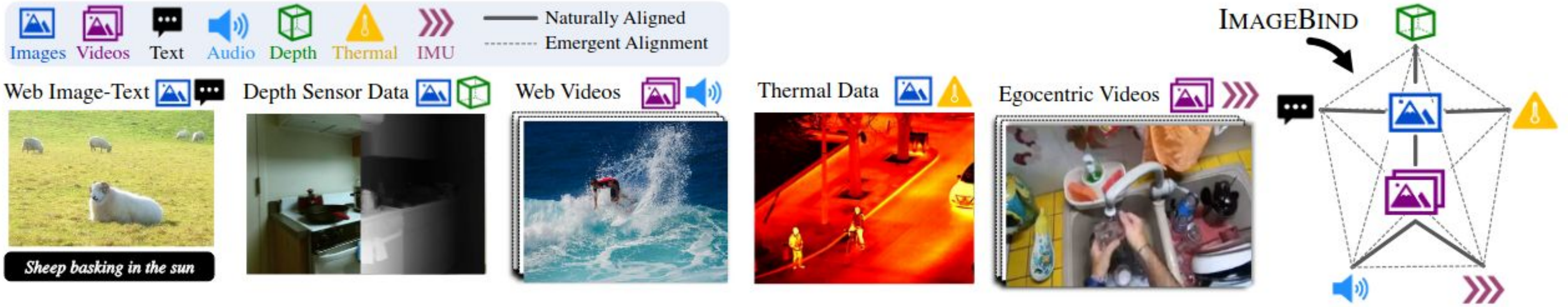
We want to detect this misalignment!

- Notify a user if their video/audio is misaligned so they can fix/repair it
- A preprocessing step so a downstream model doesn't get poisoned by data

Can we learn a joint distribution of video and audio embeddings that is representative of aligned channels, which can be used to detect poor alignment?

Related Work

ImageBind [3] → learns a joint embedding across 6 data modalities (images, text, audio, depth, thermal & IMU data)



[3] Girdhar, Rohit, et al. "Imagebind: One embedding space to bind them all." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2023.

Dataset

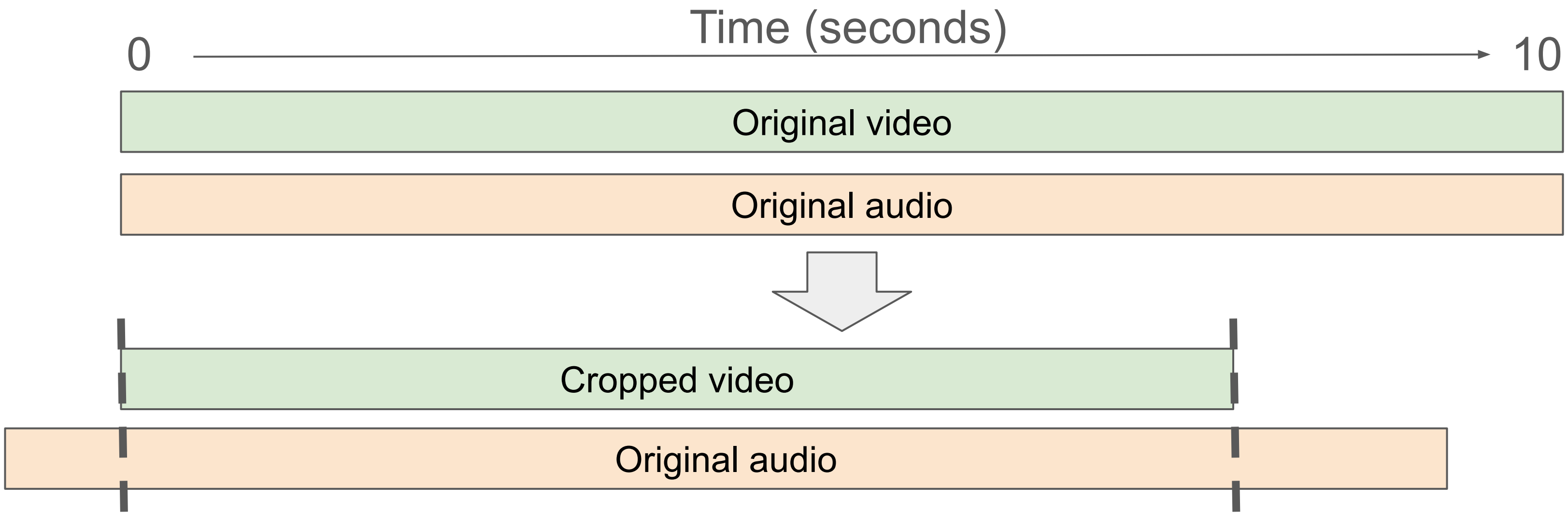
VS13 dataset:

- a data set containing on-road audio and video recordings of vehicles passing by the camera at constant speeds
- Contains 13 cars traveling at various speeds (30–105 kph)
- 400 video and audio recordings with length of 10 seconds each

[6] Djukanovic, Slobodan, et al. “A dataset for audio-video based vehicle speed estimation”. https://slobodan.ucg.ac.me/science/vs13/

Dataset Processing

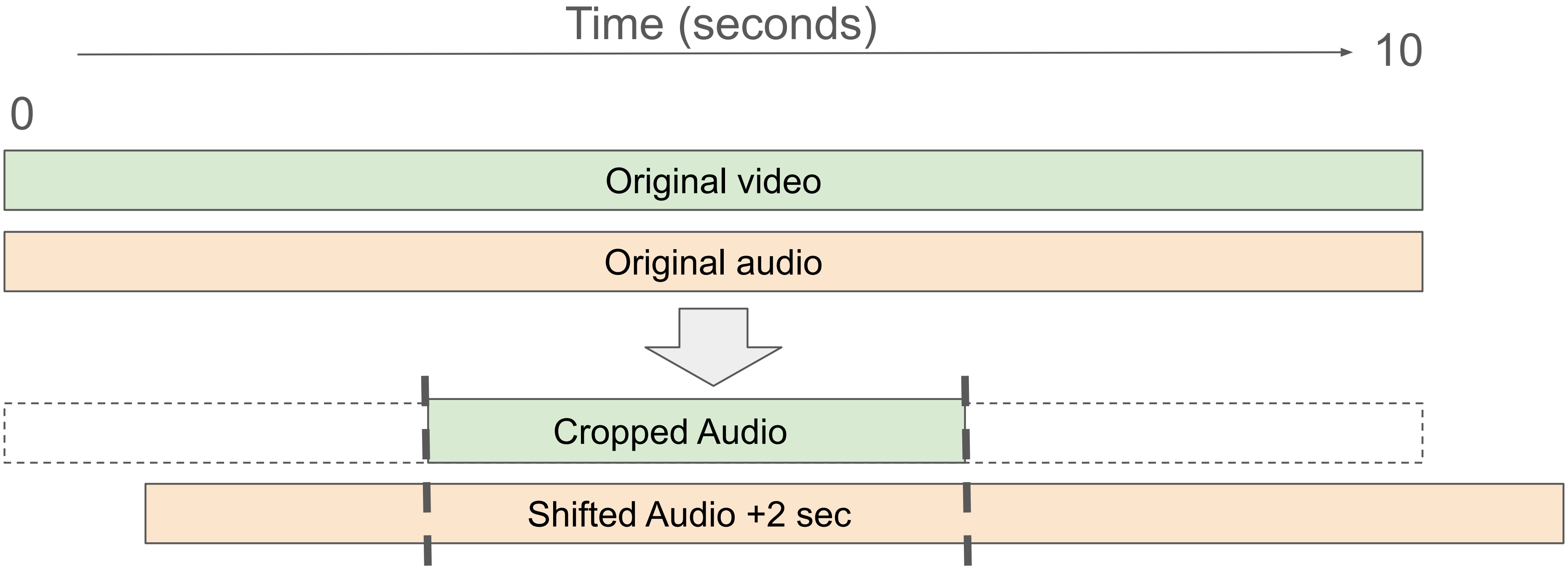
- We modified our dataset by keeping the video start time the same, and shifting the audio + or - 2 seconds
- To avoid adding artificial noise or silence from the audio offset, we cropped each video from 10 seconds to 7 seconds.



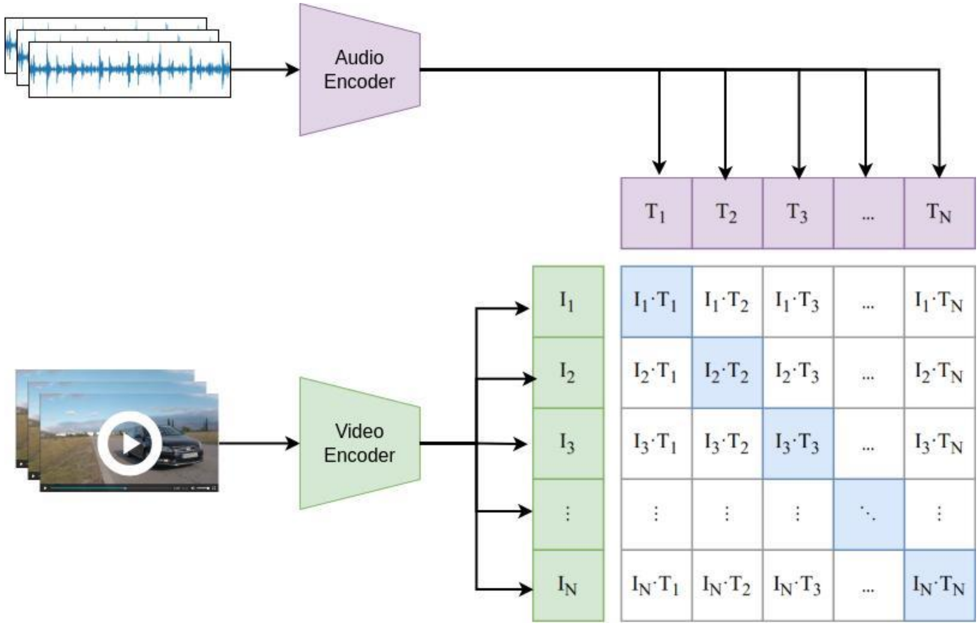
Overview: 399 total pairs, 7 seconds per clip, rough 50/50 aligned/unaligned split

Dataset Processing

- We modified our dataset by cropping the video start time to 4 seconds, and then shifting the audio + or - 2 seconds
- After processing each video and audio file were 4 seconds in length



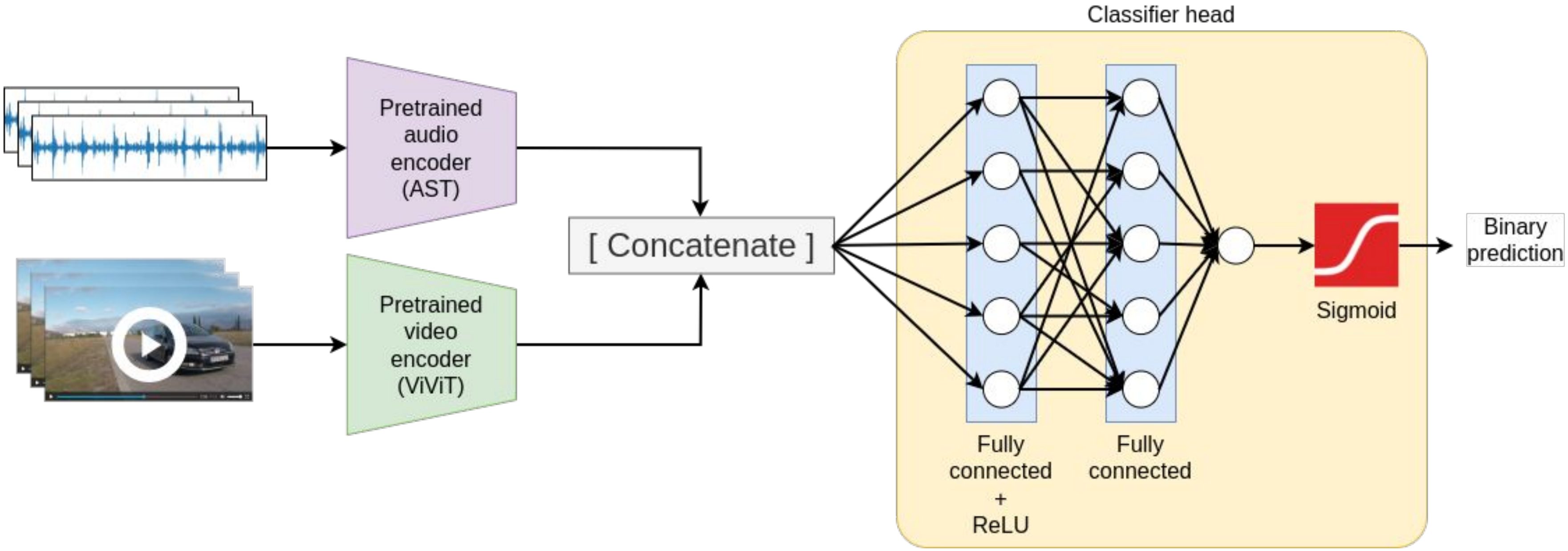
Methodology: Multi-modal transformer



⟶Learns the joint embedding of both {audio & video} data [4]

[4] Adapted from: Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PmLR, 2021.

Methodology: Multi-modal transformer



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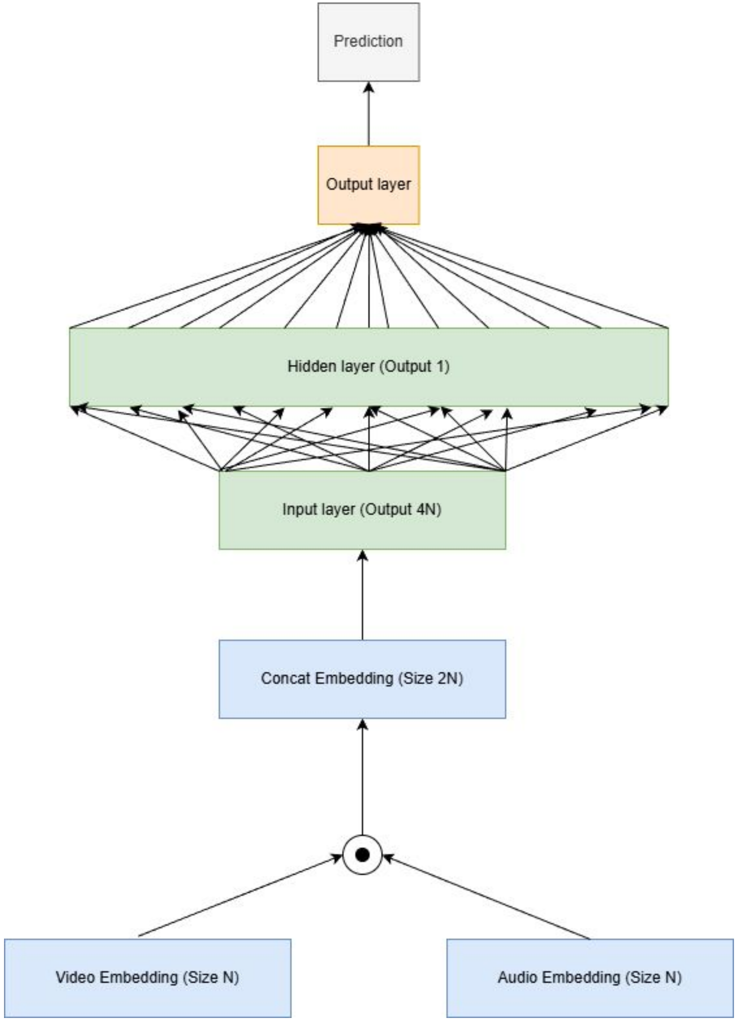


Prediction
Output layer
Hidden layer (Output 1)
Input layer (Output 4N)
Concat Embedding (Size 2N)
Video Embedding (Size N)
Audio Embedding (Size N)

Methodology: Loss Function for classification task

$$\mathcal{L} = \textbf{BCE(logits)}$$

- Intuition: We can determine if a video and audio embedding is aligned with Binary Cross Entropy (BCE) loss
  - If aligned, then embeddings should be close
  - If not aligned, then we get a higher loss
    - Backprop this loss, make the embeddings further away in this case
- Ultimate goal: make it so that our embeddings are informed about what pairs are aligned/what are not, adjust the embeddings as such

Methodology: Model Training

- **Split:** 70% training, 15% evaluation, 15% test
- **Optimizer:** Adam
- **Epochs:** 20
- **Batch size:** 50

Random hyperparameter search for:

- Number of hidden layers in transformers
- Number of attention heads
- Intermediate representation size
- Hidden representation size
- Learning rate

Hyperparameter	Selected Value
Number of hidden layers	6
Number of attention heads	3
Intermediate representation size	300
Hidden representation size	300
Learning rate	0.0001

TABLE I: Our final selection of hyperparameters after 50 steps of random hyperparamter search.

Implementation Detail: Batching

How do we batch our examples?

- Naively, load each video/audio files into vectors and preprocess them first
- Batch the processed vectors
- Issue: out of memory!

Solution: batch and perform ad hoc preprocessing

- Batch only the paths to video/audio files
- On each batch, preprocess only video/audio files in batch and then forward
- Still limited on batch size, but now we are more flexible (can test)

Evaluation

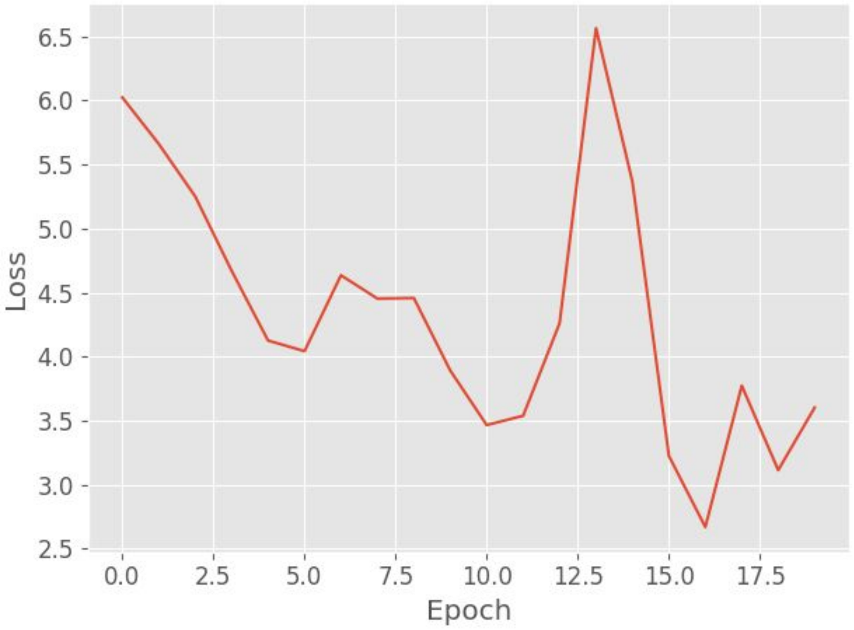
Binary classification metrics:

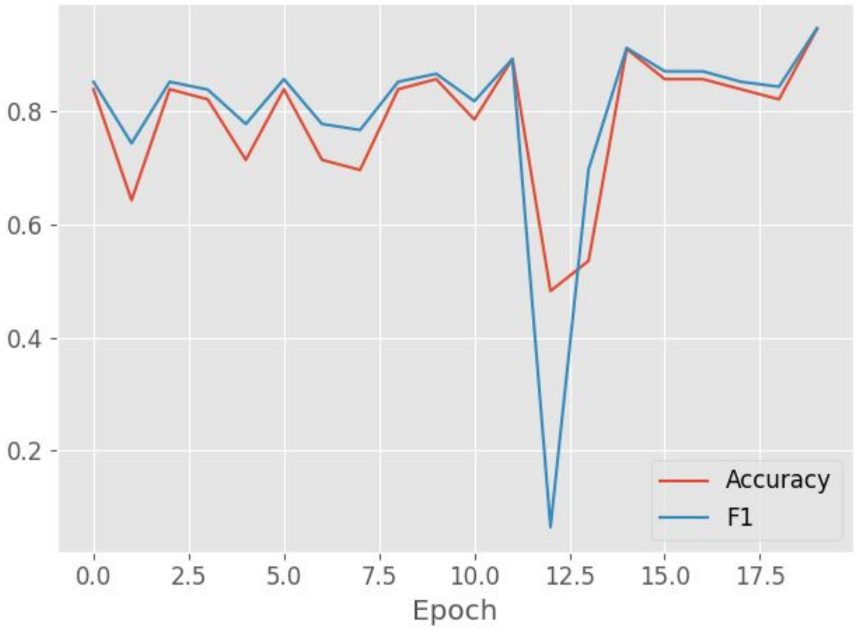
- Accuracy
  - Our dataset is balanced so as long as test set is stratified the same, this is a good metric
- Recall
- Precision
- F1-Score

Results and Insights

Model configuration	Accuracy	Precision	Recall	F1
Base model	0.786	0.8	0.667	0.727
Hidden size=120	0.571	0.602	0.532	0.565
Hidden size=600	0.375	0.375	1	0.545
3-layer classifier head	0.536	0.536	1	0.698
1-layer classifier head	0.732	0.655	0.792	0.717

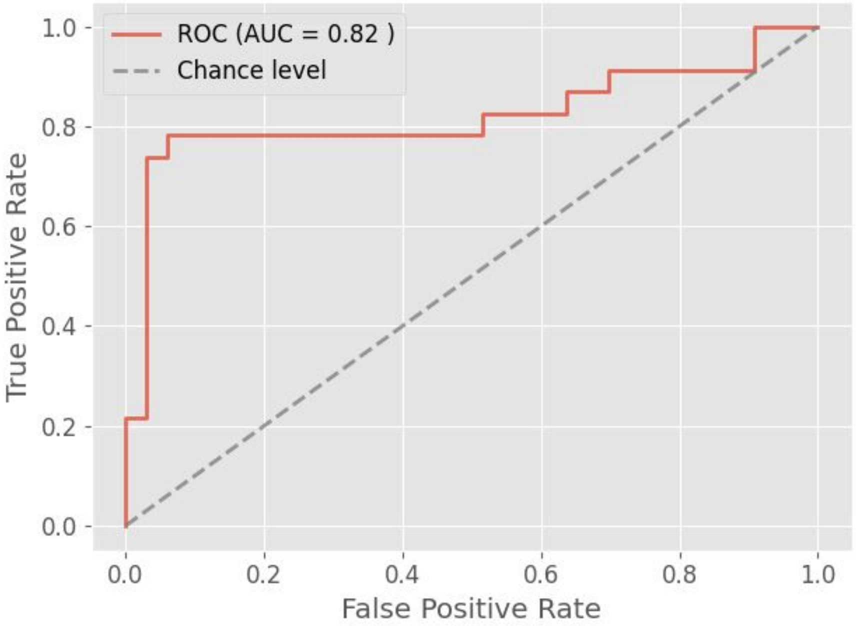
Training results

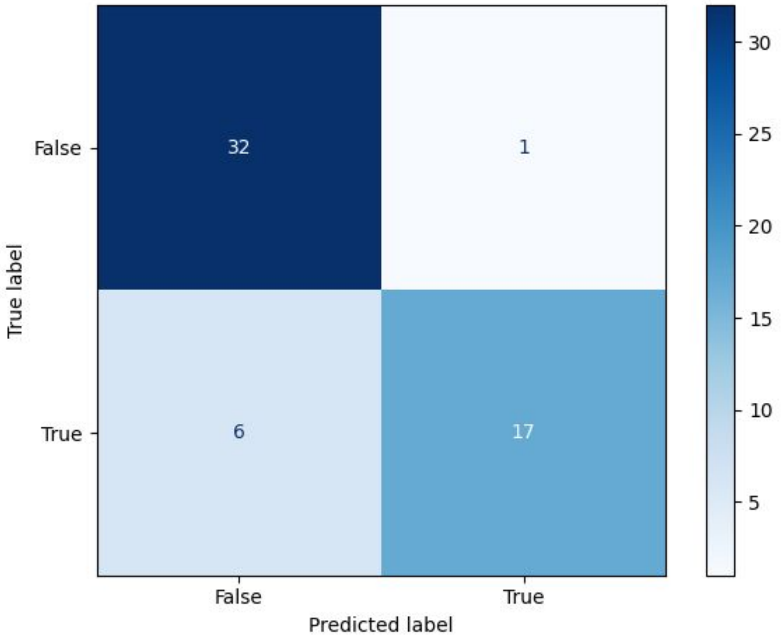




- Increase of the loss at epoch 13, but the trend is decreasing overall
- Would need more epochs to see convergence, but training was already time-consuming
- Increasing trend in accuracy and F1 score on the validation set, which is already pretty high at the first epoch due to pretraining

Results and Insights





- ROC curve plateaus in the middle and could probably be improved with more epochs
- Only 1 false positive and 6 false negative predictions on the test set

Results and Insights

We are able to successfully identify aligned/misaligned video-audio pairs with our base model!

Why does it work? Two reasons

- Quality of video and audio transformer representations of the video and audio tracks
  - ViViT and AST are able to utilize our small dataset because of its use of pre-trained image networks
  - Contextual tokens
- Fully connected classification head learns the relationship between pairs of values in the hidden representation
  - Due to our deliberate selection of the architecture

Ablation Study

We perform two ablation studies to show that our choice of FC architecture/hidden size is important

1. Retrain new models with half hidden size and double hidden size
2. Retrain new models with three layer head and one layer head

How does this affect the classification performance of the model on our target task?

Ablation Study 1

Adjusting hidden layer size

Model configuration	Accuracy	Precision	Recall	F1
Base model	0.786	0.8	0.667	0.727
Hidden size=120	0.571	0.602	0.532	0.565
Hidden size=600	0.375	0.375	1	0.545

Ablation Study 1

Both increasing and decreasing hidden representation size causes a decline in model performance

For half hidden size:

- Model not embedding enough information/features about sequences for classification head to learn meaningful relationships

For double hidden size:

- Larger embedding space for the classification head to work
- Sparsity and noise in the higher dimension space is hard to work with, classification head performance declines sharply!

Ablation Study 2

Adjusting the depth of classification head

Model configuration	Accuracy	Precision	Recall	F1
Base model	0.786	0.8	0.667	0.727
3-layer classifier head	0.536	0.536	1	0.698
1-layer classifier head	0.732	0.655	0.792	0.717

Ablation Study 2

Again both increasing and decreasing depth of classification head causes drop in performance

- Drop is less drastic than adjusting hidden size

For 1 layer:

- We don’t have neurons that are looking at the relationships of all value pairs of the video and audio embeddings, so we lose important parameters

For 3 layer:

- Too deep networks forces additional gradient flow and dampening of signal toward earlier layers
- Earlier layers are important for learning relationship between pairs!

Conclusion

- We successfully learned a joint distribution of video and audio embeddings that captures the characteristics of aligned multimodal data, enabling misalignment detection with nearly 80 percent accuracy
- Reliable real-world misalignment detection is both feasible and practically valuable

Future Work:

- Further refine the model architecture to improve performance.
- Extend the framework to additional modalities (e.g., IMU signals, text, depth)
- Train on broader categories of video-audio pairs, possibly including human speech and conversational scenarios

References

[1][2] Katharina Buchholz "YouTube Responsible For 16% Of Global Internet Traffic". https://www.forbes.com/sites/katharinabuchholz/2025/04/25/youtube-responsible-for-16-of-global-internet-traffic/

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[4] Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PmLR, 2021.

[5] Mario Parreño "ViT - The Vision Transformer". https://aidventure.es/blog/vit/

[6] Djukanovic, Slobodan, et al. "A dataset for audio-video based vehicle speed estimation". https://slobodan.ucg.ac.me/science/vs13/